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CATHODE ACTIVE MATERIAL FOR LITHIUM SECONDARY BATTERY, AND LITHIUM SECONDARY BATTERY COMPRISING SAME

Abstract

The present exemplary embodiments relate to a positive electrode active material for a lithium secondary battery, and a lithium secondary battery including the same. The positive electrode active material for a lithium secondary battery according to an exemplary embodiment includes: a metal oxide in the form of a single particle; and a coating layer positioned on the surface of the metal oxide, wherein the coating layer includes a plurality of stripe shapes including a protruding portion based on a cross section in a length direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Bypass Continuation-in-Part of International Application No. PCT/KR2023/020020 filed Dec. 6, 2023, which claims priority based on Korean Patent Application No. 10-2022-0183302 filed Dec. 23, 2022, the respective disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present exemplary embodiments relate to a positive electrode active material for a lithium secondary battery and a lithium secondary battery including the same.

BACKGROUND ART

[0003] In recent years, due to an explosive demand for electric vehicles and a demand for increased driving distance, development of a secondary battery having a high capacity and a high energy density for meeting the demands is actively progressing worldwide.

[0004] As a way to meet the demands, a technology of using a high nickel-based nickel cobalt manganese (NCM) positive electrode material having a high Ni content has been suggested. In addition, in order to improve a plate density of an electrode which is a cell constituent element, it should be composed of a bimodal form in which large particles and small particles are blended in a certain fraction.

[0005] However, since a positive electrode material form composed of a secondary particle form formed by agglomeration of primary particles having a size from tens of nm to several μm has a large powder specific surface area, it has a large area in contact with an electrolytic solution, which causes a high possibility of gassing and deterioration of life characteristics.

[0006] In order to solve the problem, a method of increasing the size of primary particles using a sintering agent, a flux, or the like has been suggested. However, in this case, a rocksalt structure is produced on the surface portion of the particles to deteriorate the electrochemical performance of the positive electrode active material.

[0007] Accordingly, development of a positive electrode active material which has excellent electrochemical performance while increasing the size of the primary particles is required.

DISCLOSURE

Technical Problem

[0008] The present disclosure attempts to provide a positive electrode active material capable of having a single particle form and having excellent electrochemical performance, and a lithium secondary battery including the same.

Technical Solution

[0009] In one general aspect, a positive electrode active material for a lithium secondary battery includes: a metal oxide in the form of a single particle; and a coating layer positioned on the surface of the metal oxide, wherein the coating layer includes a plurality of stripe shapes including a protruding portion based on a cross section in a length direction.

[0010] In another general aspect, a lithium secondary battery includes the positive electrode; a negative electrode; and an electrolyte.

Advantageous Effects

[0011] The positive electrode active material for a lithium secondary battery according to an exemplary embodiment may implement a lithium secondary battery having excellent electrochemical performance, by modifying a surface structure so that a coating layer which is in the form of a single particle and includes a stripe shape is positioned on the surface.

Description

DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is an SEM image measured at 20,000× magnification for a positive electrode active material prepared according to Example 4.

[0013] FIG. 2 is an SEM image measured at 20,000× magnification for a positive electrode active material prepared according to Comparative Example 3.

[0014] FIG. 3 is an SEM image measured at 120,000× magnification for a positive electrode active material prepared according to Example 4.

[0015] FIG. 4 is an SEM image measured at 60,000× magnification for a vertical cross section after processing a cross section in a length direction (direction B in FIG. 4) in the positive electrode active material prepared according to Example 4 using a focused ion beam (FIB).

[0016] FIG. 5 is schematically shown in order to describe a distance and a height between protruding portions forming a stripe shape of area A and area B in FIG. 4.

[0017] FIG. 6 is an SEM image measured at 5,000× magnification for a positive electrode active material prepared according to Example 8.

[0018] FIG. 7 shows results of measuring HPPC resistance values of positive electrode active materials prepared according to Example 4 and Comparative Example 3.

MODE FOR INVENTION

[0019] The terms such as first, second, and third are used for describing various parts, components, areas, layers, and/or sections, but are not limited thereto. These terms are used only for distinguishing one part, component, area, layer, or section from other parts, components, areas, layers, or sections. Therefore, a first part, component, area, layer, or section described below may be mentioned as a second part, component, area, layer, or section without departing from the scope of the present disclosure.

[0020] The terminology used herein is only for mentioning a certain example, and is not intended to limit the present disclosure. Singular forms used herein also include plural forms unless otherwise stated clearly to the contrary. The meaning of “comprising” used in the specification is embodying certain characteristics, areas, integers, steps, operations, elements, and/or components, but is not excluding the presence or addition of other characteristics, areas, integers, steps, operations, elements, and/or components.

[0021] When it is mentioned that a part is “on” or “above” the other part, it means that the part is directly on or above the other part or another part may be interposed therebetween. In contrast, when it is mentioned that a part is “directly on” the other part, it means that nothing is interposed therebetween.

[0022] Though not defined otherwise, all terms including technical terms and scientific terms used herein have the same meaning as commonly understood by a person with ordinary skill in the art to which the present disclosure pertains. Terms defined in commonly used dictionaries are further interpreted as having a meaning consistent with the related technical literatures and the currently disclosed description, and unless otherwise defined, they are not interpreted as having an ideal or very formal meaning.

[0023] As described above, when the size of primary particles is increased, a rocksalt structure is produced on the surface to deteriorate electrochemical performance of a positive electrode active material. However, in the present exemplary embodiment, the problem has been solved by implementing a positive electrode active material having a modified surface structure so that a coating layer including a stripe-shaped pattern including a protruding portion is positioned on the surface of a metal oxide in the form of a single particle with an increased size of primary particles.

[0024] That is, the positive electrode active material for a lithium secondary battery according to an exemplary embodiment includes: a metal oxide in the form of a single particle, and a coating layer positioned on the surface of the metal oxide, wherein the coating layer includes a plurality of stripe shapes including a protruding portion based on a cross section in a length direction.

[0025] Herein, an average length in a short side direction of the protruding portion forming the stripe shape included in the coating layer may be in a range of 110 nm or more, more specifically 110 nm to 250 nm.

[0026] In addition, an average length in a long side direction forming the stripe shape included in the coating layer may be in a range of 650 nm or more, more specifically 680 nm to 1500 nm.

[0027] When a distance between the protruding portions and an average height of the protruding portions included in the stripe shapes formed on the coating layer satisfy the above range, the electrochemical performance of a lithium secondary battery to which the positive electrode active material according to the present exemplary embodiment is applied may be dramatically improved.

[0028] Next, the average distance between the protruding portions forming the stripe shapes included in the coating layer may be in a range of 10 nm to 60 nm or 18 nm to 45 nm. In addition, the average height of the protruding portions may be in a range of 18 nm to 75 nm or 25 nm to 65 nm.

[0029] The stripe shape may satisfy the following Equation 1:

[00001] $4 < B / A < 11$ [Equation1] [0030] wherein A is a length in a short side direction of the protruding portion forming the stripe shape, and B is a length in a long side direction of the protruding portion forming the stripe shape.

[0031] Specifically, Equation 1 may be in a range of 5 to 10. When lengths in width and length directions of the protruding portion forming the stripe shape satisfy the range of Equation 1, a positive electrode active material which has a significantly decreased high-temperature resistance increase rate while having excellent room temperature resistance and high-temperature life characteristics may be provided.

[0032] In addition, the stripe shape may satisfy the following Equation 2:

[00002] $0.1 < D / C < 8$ [Equation2] [0033] wherein C is a distance between the protruding portions forming the stripe shape, and D is a height of the protruding portion forming the stripe.

[0034] Specifically, Equation 2 may be in a range of 0.5 to 5 or 0.7 to 4. When a ratio between the distance between the protruding portions and the height of the protruding portion, forming the stripe shape, satisfies the range of Equation 2, a positive electrode active material which has a significantly decreased high-temperature resistance increase rate while having excellent room temperature resistance and high-temperature life characteristics.

[0035] Meanwhile, the positive electrode active material may have a Li/Ni cation mixing ratio of 1.5% or less, more specifically 1.1 to 1.4%. When the Li/Ni cation mixing ratio is too large, a Li layer may easily collapse, which may greatly decrease the life characteristics of a battery. In addition, the Li/Ni cation mixing ratio is too small, an irreversible site of a bulk portion of the positive electrode active material may be increased to reduce lithium ion mobility, which may deteriorate resistance characteristics and output characteristics. Accordingly, when the Li/Ni cation mixing ratio satisfies the range, a positive electrode active material having low resistance and improved lifespan may be implemented, and thus, has an advantageous effect.

[0036] In the present specification, the Li/Ni cation mixing ratio refers to an amount of Ni substituted on a Li site.

[0037] In the present exemplary embodiment, the coating layer may include at least one of Co, Al, W, V, Ti, Nb, Ce, B, and P. Herein, a content of the element included in the coating layer may be in a range of 0.5 mol % to 3.5 mol %, more specifically 1.0 mol % to 3.0 mol %, or 1.5 mol % to 2.5 mol %, based on the entire coating layer. Since the coating layer includes at least one of the elements described above within the range described above, the surface structure of the positive electrode active material of the present exemplary embodiment may be modified.

[0038] The metal oxide includes nickel, cobalt, and manganese, and the content of nickel may be higher than the sum of the contents of cobalt and manganese in the entire metal oxide.

[0039] More specifically, the content of nickel in the metal oxide particles may be 0.8 mol or more based on 1 mol of nickel, cobalt, and manganese. More specifically, the content of nickel may be in a range of 0.8 to 0.99, 0.85 to 0.99, or 0.88 to 0.99 range.

[0040] When the content of nickel in the metals of the lithium metal oxide is 0.8 mol or more, a positive electrode active material having high output characteristics may be implemented. Since the positive electrode active material of the present exemplary embodiment having the composition has an increased energy density per volume, the capacity of a battery to which the positive electrode active material is applied may be improved and the positive electrode active material is very appropriate for use as an electric vehicle.

[0041] The metal oxide further includes a doping element, and the doping element may include at least one of Al, Zr, Nb, Mo, W, Ti, Ce, Mg, B, P, V, Sr, and B.

[0042] The content of the doping element may be in a range of 0.0005 mol to 0.04 mol or 0.001 mol to 0.03 mol, based on a total of 1 mol of nickel, cobalt, manganese, and the doping elements. Herein, the doping element refers to a doping amount of the doping elements included in a finally obtained positive electrode active material.

[0043] In the positive electrode active material, selection of the doping element is important for securing lifespan and various electrochemical performances. In the present exemplary embodiment, the positive electrode active material characteristics may be improved by applying various doping elements as described above.

[0044] In the present exemplary embodiment, the doping element may include Zr and Al.

[0045] Since a Zr ion occupies a Li site, Zr serves as a type of pillar and relieves contraction of a lithium ion path during charging and discharging process to stabilize a layered structure. The phenomenon may decrease so-called cation mixing and increase a lithium diffusion coefficient to increase cycle life.

[0046] In addition, an Al ion moves to a tetragonal lattice site to suppress deterioration of the layered structure into a spinel structure in which movement of a lithium ion is less smooth.

[0047] The content of Zr may be in a range of 0.0005 mol to 0.01 mol, more specifically, 0.0005 to 0.005 mol, 0.0005 mol to 0.003 mol, or 0.001 mol to 0.0025 mol, based on a total of 1 mol of nickel, cobalt, manganese, and the doping elements.

[0048] When the Zr doping amount satisfies the range, a high-temperature resistance increase rate may be decreased, and simultaneously excellent life characteristics may be secured.

[0049] The content of Al may be in a range of 0.0005 mol to 0.04 mol, more specifically 0.0005 mol to 0.028 mol, 0.001 mol to 0.005 mol, or 0.002 mol to 0.004 mol, based on a total of 1 mol of nickel, cobalt, manganese, and the doping elements.

[0050] When the Al doping amount satisfies the range, high-temperature lifespan and thermal stability may be further improved.

[0051] Next, an average crystalline size of the metal oxide may be 200 nm or more.

[0052] When the metal oxide has the average crystalline size, it may be defined as a single particle. In addition, when the average crystalline size satisfies the range, crystallization is performed well, so that residual lithium on the surface of the positive electrode active material may be reduced, and

the life characteristics of the lithium secondary battery may be further improved.

[0053] In the present specification, the average crystalline size is defined as the size measured by the following method: [0054] 1. Sample structure analysis using Rigaku smart lab equipment [0055] 2. Applying 45 kV, 200 mA (9 kW) to a Cu anode in order to generate X-ray [0056] 3. Setting equipment optic to an incident slit of $\frac{1}{2}$ deg and a receiving slit of 8.0 mm [0057] 4. Performing XRD measurement at scan of 10-80°, step of 0.02°, 10°/min [0058] 5. Calculation of crystal size using SmartLab Studio II v4.2.82.0 S/W [0059] 6. Using Whole powder pattern fitting (WPPF) in S/W for calculation [0060] 7. During WPPF, using a layered structure as the structure and setting profile fitting by a FP method [0061] 8. Setting receiving optic to Graphite 002 and Soller slit of 3.8 [0062] 9. Calculating a crystal size by refining shape to spherical and strain to 0 Fix

[0063] An average particle diameter (D50) of the positive electrode active material may be in a range of 2.5 μm or more, more specifically 3.0 μm to 5.0 μm . In the present exemplary embodiment, a positive electrode active material in the form of a single particle having a uniform particle size distribution which has very little fine powder and large powder, without separate expensive shredding equipment or multiple shredding processes, that is, even using a common shredding device, in order to prepare a positive electrode active material in the form of a single particle having the average particle diameter described above, may be prepared. Accordingly, when the average particle diameter of the positive electrode active material of the present exemplary embodiment satisfies the range, a lithium secondary battery having excellent electrochemical properties may be implemented.

[0064] Meanwhile, the positive electrode active material of the present exemplary embodiment may further include a positive electrode active material including a metal oxide in the form of a secondary particle formed by agglomeration of primary particles.

[0065] That is, the positive electrode active material including the metal oxide in the form of a single particle, and a positive electrode active material including a metal oxide in the form of a secondary particle having an average particle diameter (D50) larger than an average particle diameter (D50) of the positive electrode active material including the metal oxide in the form of a single particle may be included. As such, when the positive electrode active material including the metal oxide in the form of a single particle and the positive electrode active material including the metal oxide in the form of a secondary particle are mixed in a bimodal form as described above, an electrode mixed density may be increased, which is thus preferred.

[0066] As such, a mixing ratio between the positive electrode active material including the metal oxide in the form of a single particle and the positive electrode active material including the metal oxide in the form of a secondary particle in the positive electrode active material in a bimodal form may be in a range of 30:70 to 10:90 or 25:75 to 15:85, as a weight ratio (single particle:primary particle). When the positive electrode active materials including the metal oxides in the form of a single particle and in the form of a secondary particle are mixed at the weight ratio, an electrode mixed density may be increased.

[0067] Herein, the positive electrode active materials including the metal oxides in the form of a single particle and in the form of a secondary particle may have the same composition as or different compositions from each other. Specifically, both the positive electrode active materials including the metal oxides in the form of a single particle and in the form of a secondary particle may include nickel, cobalt, manganese, and doping elements.

[0068] For example, the positive electrode active material including the metal oxide in the form of a secondary particle may include nickel, cobalt, and manganese, and the content of nickel may be higher than the sum of the contents of cobalt and manganese in the entire metal oxide in the form of a secondary particle.

[0069] More specifically, the content of nickel in the metal oxide particles in the form of a secondary particle may be 0.8 mol or more based on 1 mol of nickel, cobalt, and manganese. More

specifically, the content of nickel may be in a range of 0.8 to 0.99, 0.85 to 0.99, or 0.88 to 0.99.

[0070] The metal oxide in the form of a secondary particle may further include a doping element, and the doping element may include at least one of Al, Zr, Nb, Mo, W, Ti, Ce, Mg, B, P, V, Sr, and B.

[0071] The detailed description and the contents of the doping elements will be omitted, since they are the same as those of the positive electrode active material including the metal oxide in the form of a single particle described above.

[0072] In addition, in the present exemplary embodiment, the average particle diameter (D50) of the positive electrode active material including the metal oxide in the form of a secondary particle may be in a range of 10 μm to 20 μm or 12 μm to 17 μm . When the average particle diameter of the positive electrode active material including the metal oxide in the form of a secondary particle satisfies the range, large and small particles may be positioned in an appropriate distribution in the positive electrode active material in a bimodal form, and thus, the energy density of a lithium secondary battery may be improved.

Lithium Secondary Battery

[0073] Another exemplary embodiment of the present disclosure provides a positive electrode including the positive electrode active material according to an exemplary embodiment of the present disclosure described above, a negative electrode, and an electrolyte positioned between the positive electrode and the negative electrode.

[0074] Specifically, the positive electrode includes a current collector and a positive electrode active material layer formed on the current collector, and the characteristics of the positive electrode active material forming the positive electrode active material layer are as described above. Accordingly, the detailed description of the positive electrode active material will be omitted.

[0075] As the current collector, for example, stainless steel, aluminum, nickel, titanium, calcined carbon, or aluminum or stainless steel having a surface treated with carbon, nickel, titanium, silver, and the like may be used.

[0076] Meanwhile, the positive electrode active material layer may include a binder and a conductive material.

[0077] The binder serves to adhere positive electrode active material particles to each other well and adhere the positive electrode active material to the current collector well.

[0078] The conductive material is used for imparting conductivity to an electrode, and in a battery to be configured, any electroconductive material without causing a chemical change may be used.

[0079] The positive electrode is manufactured by mixing an active material, a conductive material, and a binder in a solvent to prepare an active material composition, and applying the composition on a current collector. Since the method for manufacturing an electrode as such is well known in the art, the detailed description will be omitted in the present specification. As the solvent, N-methylpyrrolidone and the like may be used, but the present disclosure is not limited thereto.

[0080] The negative electrode includes a current collector and a negative electrode active material layer formed on the current collector, and the negative electrode active material layer includes a negative electrode active material.

[0081] The negative electrode active material includes a material capable of reversibly intercalating/deintercalating lithium ions, a lithium metal, an alloy of lithium metal, a material capable of doping and dedoping lithium, or a transition metal oxide.

[0082] The material capable of reversibly intercalating/deintercalating lithium ions is a carbon material and may be any carbon-based negative electrode active material which is commonly used in a lithium ion secondary battery, and a representative example may be crystalline carbon, amorphous carbon, or a combination thereof.

[0083] The alloy of lithium metal may be an alloy of lithium and a metal selected from the group consisting of Na, K, Rb, Cs, Fr, Be, Mg, Ca, Sr, Si, Sb, Pb, In, Zn, Ba, Ra, Ge, Al, and Sn.

[0084] As the material capable of doping and dedoping lithium may be Si, $\text{SiO}_{0.5x}$ ($0 < x < 2$), a Si—Y alloy (Y is an element selected from the group consisting of alkali metals, alkaline earth metals, Group 13 elements, Group 14 elements, transition metals, rare earth elements, combinations thereof, but is not Si), Sn, $\text{SnO}_{0.5}$, Sn—Y (Y is an element selected from the group consisting of alkali metals, alkaline earth metals, Group 13 elements, Group 14 elements, transition metals, rare earth elements, combinations thereof, but is not Sn,) and the like.

[0085] The transition metal oxide may be a vanadium oxide, a lithium vanadium oxide, and the like.

[0086] The negative electrode active material layer may also include a binder, and optionally, further include a conductive material.

[0087] As the binder, polyvinyl alcohol, carboxymethyl cellulose/styrene-butadiene rubber, hydroxypropylene cellulose, diacetylene cellulose, polyvinyl chloride, polyvinylpyrrolidone, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, or the like may be used, but the present disclosure is not limited thereto. The binder may be mixed at 1 wt % to 30 wt % with respect to the total amount of a composition for forming the negative electrode active material layer.

[0088] The conductive material is not particularly limited as long as it has conductivity without causing a chemical change in a battery, and specifically, may be graphite such as natural graphite and artificial graphite; carbon black such as acetylene black, Ketjen black, channel black, furnace black, lamp black, and thermal black; conductive fiber such as carbon fiber and metal fiber; metal powder such as carbon fluoride, aluminum, and nickel powder; conductive whisker such as zinc oxide and potassium titanate; a conductive metal oxide such as titanium oxide; a conductive material such as a polyphenylene derivative, and the like. The conductive material may be mixed at 0.1 wt % to 30 wt % with respect to the total weight of the composition for a negative electrode active material layer.

[0089] As the current collector, a current collector selected from the group consisting of a copper foil, a nickel foil, a stainless steel foil, a titanium foil, a nickel foam, a copper foam, a polymer substrate coated with a conductive metal, and a combination thereof may be used.

[0090] The negative electrode is manufactured by mixing an active material, a conductive material, and a binder in a solvent to prepare an active material composition, and applying the composition on a current collector. Since the method for manufacturing an electrode as such is well known in the art, the detailed description will be omitted in the present specification. As the solvent, N-methylpyrrolidone and the like may be used, but the present disclosure is not limited thereto.

[0091] As the electrolyte, an organic liquid electrolyte, an inorganic liquid electrolyte, a solid polymer electrolyte, a gel polymer electrolyte, a solid inorganic electrolyte, a molten type inorganic electrolyte, and the like which may be used in the manufacture of a lithium secondary battery may be used, but the present disclosure is not limited thereto.

[0092] Specifically, the organic liquid electrolyte includes a nonaqueous organic solvent and a lithium salt.

[0093] The nonaqueous organic solvent serves as a medium in which ions involved in an electrochemical reaction of a battery may move.

[0094] The lithium salt is a material which is dissolved in an organic solvent, acts as a source of a lithium ion in a battery to allow basic operation of a lithium secondary battery, and serves to promote movement of lithium ions between a positive electrode and a negative electrode.

[0095] There may be a separator between the positive electrode and the negative electrode, depending on the type of lithium secondary batteries. As the separator, polyethylene, polypropylene, polyvinylidene fluoride, or a multilayer membrane of two or more layers thereof may be used, and a mixed multilayer membrane such as a polyethylene/polypropylene 2-layer separator, a polyethylene/polypropylene/polyethylene 3-layer separator, and a polypropylene/polyethylene/polypropylene 3-layer separator may be used, of course.

[0096] The lithium secondary battery may be classified into a lithium ion battery, a lithium ion polymer battery, and a lithium polymer battery, depending on the type of separators and electrolytes to be used, may be classified into a cylindrical type, a prismatic type, a coin type, a pouch type, and the like, depending on the shape, and may be classified into a bulk type and a thin membrane type, depending on the size. Since the structure and the manufacturing method of the battery are well known in the art, the detailed description thereof will be omitted.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0097] Hereinafter, exemplary embodiments of the present disclosure will be described in detail. However, these are suggested only as an example and the present disclosure is not limited thereby, and the present disclosure is only defined by the scope of the claims described later.

Example 1—Positive Electrode Active Material in the Form of Single Particle, Heat Treatment at 600° C. For 6 hr

(1) Preparation of Precursor

[0098] A precursor was prepared by a common co-precipitation method.

[0099] Specifically, $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ was used as a nickel raw material, $\text{CoSO}_4 \cdot 7\text{H}_2\text{O}$ was used as a cobalt raw material, and $\text{MnSO}_4 \cdot \text{H}_2\text{O}$ was used as a manganese raw material. These raw materials were dissolved in distilled water to prepare a metal salt aqueous solution.

[0100] After preparing a co-precipitation reactor, N_2 was purged in order to prevent oxidation of a metal ion in a co-precipitation reaction, and a temperature of the reactor was maintained at 50° C.

[0101] NH_4OH as a chelating agent was added to the co-precipitation reactor, and NaOH was used for pH adjustment. Precipitate obtained according to the co-precipitation process was filtered, washed with distilled water, and dried in an oven at 100° C. for 24 hours to prepare a positive electrode active material precursor.

[0102] The composition of the prepared precursor was $(\text{Ni}_{0.98}\text{Co}_{0.01}\text{Mn}_{0.01})(\text{OH})_2$, and the average particle diameter (D_{50}) thereof was about 4 μm .

(2) Preparation of Positive Electrode Active Material

[0103] 315 g of $\text{LiOH} \cdot \text{H}_2\text{O}$, 2.94 g of $\text{Al}(\text{OH})_3$, and 1.47 g of ZrO_2 were weighed, respectively, in 684 g of the precursor prepared in (1), and then uniformly mixed, and fired at 830-890° C. for 24 hours in a firing furnace in a box form to which oxygen was introduced at 1,000 mL/min, thereby synthesizing a positive electrode material in a metal oxide form. Thereafter, the positive electrode material was crushed using a jet-mill to prepare a positive electrode active material including a metal oxide in the form of a single particle.

[0104] About 1.927 g of $\text{Co}(\text{OH})_2$ was mixed per 100 of the metal oxide in the form of a single particle and a heat treatment was performed at 600° C. for 6 hours under an oxygen atmosphere to prepare a positive electrode active material having a coating layer formed thereon.

Examples 2 to 7

[0105] Positive electrode active materials having a coating layer formed thereon were prepared in the same manner as in Example 1, except that heat treatment conditions during coating layer formation were adjusted as shown in the following Table 1.

TABLE-US-00001	TABLE 1	Coating layer forming process	Heat treatment	Heat treatment
	Classification	temperature (° C.)	time (hr)	
Example 1	600	6	Example 2	620
Example 3	640	6	Example 4	660
Example 5	680	6	Example 6	700
Example 7	720	6		

Example 8 (1) Preparation of Precursor

[0106] A precursor was prepared in the same manner as in Example 1, except that the composition was $(\text{Ni}_{0.92}\text{Co}_{0.04}\text{Mn}_{0.04})(\text{OH})_2$, and the average particle diameter (D_{50}) was about 14.5 μm .

(2) Preparation of Positive Electrode Active Material Having Large Particle Diameter

[0107] 325 g of $\text{LiOH} \cdot \text{H}_2\text{O}$, 11.4 g of $\text{Al}(\text{OH})_3$, and 3.2 g of ZrO_2 were

weighed, respectively in 684 g of the precursor prepared in (1), uniformly mixed, and fired at 830-890° C. for 24 hours in a firing furnace in a box form to which oxygen was introduced at 1,000 mL/min, thereby synthesizing a positive electrode material in a metal oxide form.

[0108] Thereafter, the positive electrode material was crushed using an air classifying mill (ACM), washed for removing residual lithium on the surface, and dried for 12 hours.

[0109] Next, a mixture of about 0.35 g of boric acid (H_3BO_3) per 100 of the dried positive material was heat treated at 250° C.-300° C. for 5 hours under an air atmosphere.

(3) Preparation of Bimodal Positive Electrode Active Material

[0110] A bimodal positive electrode active material was prepared by mixing the positive electrode active material prepared in (2) and the positive electrode active material prepared according to Example 4 at a weight ratio of 8:2.

Comparative Example 1

(1) Preparation of Precursor

[0111] A precursor was prepared in the same manner as in Example 1.

(2) Preparation of Positive Electrode Active Material

[0112] 315 g of $\text{LiOH} \cdot \text{H}_2\text{O}$, 2.94 g of $\text{Al}(\text{OH})_3$, and 1.47 g of ZrO_2 were weighed, respectively, in 684 g of the precursor prepared in (1), uniformly mixed, and fired at 830-890° C. for 24 hours in a firing furnace in a box form to which oxygen was introduced at 1,000 mL/min, thereby synthesizing a positive electrode material in a metal oxide form. Thereafter, the positive electrode material was crushed using a jet-mill to prepare a positive electrode active material including a metal oxide in the form of a single particle.

Comparative Example 2

[0113] A positive electrode active material according to Comparative Example 2 was prepared by further heat treating the positive electrode active material prepared according to Comparative Example 1 at 660° C. for 6 hours.

Comparative Example 3—Forming Co Coating Layer in a Wet Manner

[0114] A positive electrode active material was prepared in the same manner as in Comparative Example 1.

[0115] Next, 0.03 g of LiNO_3 and 11.74 g of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ were dissolved in 100 g of H_2O heated to 60° C., a slurry obtained by adding 100 g of the positive electrode material of Comparative Example 1 was stirred for 30 minutes, and the slurry was spray dried to obtain a powder body. The powder body was placed in an alumina crucible, heated to about 450° C. at about 5° C. per minute, and maintained at about 450° C. for about 1 hours, and the temperature was heated to about 700° C. at about 2° C. per minute and then maintained for about 2 hours. Subsequently, the sample was cooled to room temperature to prepare the positive electrode active material according to Comparative Example 2.

Experimental Example 1: Evaluation of Electrochemical Performance

(1) Manufacture of Coin-Type Half Battery

[0116] In order to evaluate the physical properties and electrochemistry of the positive electrode active materials prepared in Examples 1 to 8 and Comparative Examples 1 and 2, a coin-type half battery was manufactured as follows.

[0117] Specifically, the positive electrode active material, a polyvinylidene fluoride binder (product name: KF1120), and a carbon black conductive material were mixed at a weight ratio of 96.5:1.5:2, and the mixture was added to a N-methyl-2-pyrrolidone solvent to the solid content of about 30 wt %, thereby preparing a positive electrode active material slurry.

[0118] An aluminum foil (thickness: 15 μm) as a positive electrode current collector was coated with the slurry using a doctor blade, dried, and rolled to manufacture a positive electrode. The loading amount of the positive electrode was about 15 mg/cm², an electrode thickness was about 65 μm , and a rolling density was about 3.4 g/cm³ or more.

[0119] A 2032 coin-type half battery was manufactured by a common method, using the positive

electrode, a lithium metal negative electrode (thickness: 300 μm , MTI), an electrolytic solution, and a polypropylene separator. As the electrolyte solution, a mixed solution obtained by dissolving 1M LiPF₆ in a mixed solvent of ethylene carbonate, dimethyl carbonate, and ethyl methyl carbonate (EMC) (mixing ratio of EC:DMC:EMC=3:4:3 vol %) was used. After manufacturing the half battery, it was aged at room temperature for 10 hours.

(2) Measurement of Residual Lithium and Particle Size

[0120] Residual lithium was measured using T50 model available from METTLER TOLEDO, and a particle size was measured using S3500 model available from Microtrac.

(3) Capacity Evaluation

[0121] The coin-type half battery manufactured according to experimental preparation was aged at room temperature (25° C.) for 10 hours, and then was subjected to a charge/discharge test.

[0122] For capacity evaluation, 200 mAh/g was used as a reference capacity, and CC/CV 2.5-4.25V and 1/20 C cut-off were applied as charge and discharge conditions. Initial capacity was measured at charging at 0.2 C/discharging at 0.2 C.

(4) Measurement of Life Characteristics

[0123] Life characteristics were measured 50 times at a high temperature (45° C.) in the conditions of charging at 0.5 C/discharging at 1.0 C.

(5) Measurement of Resistance Characteristics

[0124] Initial room temperature resistance (DC internal resistance: DC-IR (direct current internal resistance)) was obtained by performing charge at 0.2 C and discharge at 0.2 of a battery once under the conditions of a constant current-constant voltage of 2.5V to 4.25V and 1/20C cut-off at 25° C., applying a discharge current at 100% charge at 4.25 V, measuring a voltage value after 60 seconds, and then performing calculation.

[0125] A resistance increase rate was obtained by measuring resistance at 45° C. in the same manner as in the initial resistance measurement method, after 30 times of cycle life as compared with the initially measured resistance (initial room temperature resistance), and then converting the increase rate as a percentage (%).

Experimental Example 2: X-Ray Diffraction Evaluation

[0126] Sample structure analysis using Rigaku smart lab equipment [0127] Applying 45 kV, 200 mA (9 kW) to a Cu anode for generating X-ray [0128] Setting equipment optic to an incident slit of ½ deg and a receiving slit of 8.0 mm [0129] XRD measurement at scan of 10-80°, step of 0.02°, 10°/min

Experimental Example 3: Method for Measuring Cation Mixing Ratio

[0130] 1. Sample structure analysis using Rigaku smart lab equipment [0131] 2. Applying 45 kV, 200 mA (9 kW) to a Cu anode for generating X-ray [0132] 3. Setting equipment optic to an incident slit of ½ deg and a receiving slit of 8.0 mm [0133] 4. XRD measurement at scan of 10-80°, step of 0.02°, 10°/min [0134] 5. Calculating crystal size using SmartLab Studio II v4.2.82.0 S/W [0135] 6. Using whole powder pattern fitting (WPPF) in S/W for calculation [0136] 7. Using layered structure as the structure and setting a profile fitting by FP method in proceeding with WPPF [0137] 8. Setting receiving optic to graphite 002 and soller slit to 3.8 [0138] 9. Calculating a crystal size by refining a shape to spherical and a strain to 0 Fix [0139] 10. Using occupancy of Ni1 in a crystal structure tap as a cation mixing value

TABLE-US-00002 TABLE 2 Charge Discharge High- Particle capacity capacity High- High- temperature Residual size Crystalline Cation at at temperature temperature resistance Lithium[wt %] [D50, size mixing 0.2 C. 0.2 C. Rate resistance lifespan increase LiOH Li.sub.2CO.sub.3 Total um] [nm] (%) [mAh/g] [mAh/g] [%] [Ω] [%] rate [%] Comparative 0.81 0.32 1.13 3.4 253 1.4 249.7 212.4 85.0 46.5 90.5 45.2 Example 1 Comparative 0.18 0.60 0.78 3.7 254 1.3 248.5 214.0 86.1 45.8 91.0 40.1 Example 2 Comparative 0.10 0.14 0.24 3.8 239 1.7 246.3 214.6 87.1 28 91.8 72.8 Example 3 embodiment 1 0.13 0.50 0.63 3.8 264 1.1 248.6 214.8 86.4 45.3 91.7 38.4 Example 2 0.14 0.49 0.63 3.8 264 1.5 249.1 216.9 87.1 40.5 92.4 37.6 Example 3 0.14 0.43 0.57 3.8 257 1.4

247.4 216.6 87.6 38.2 92.4 35.6 Example 4 0.14 0.39 0.53 3.8 253 1.4 247.8 216.3 87.3 36.6 92.2 34.0 Example 6 0.14 0.38 0.53 3.8 248 1.3 247.8 215.3 86.9 34.1 91.1 37.8 Example 7 0.12 0.44 0.56 3.9 247 1.4 248.7 215.9 86.8 35.1 90.4 38.8 Example 8 0.14 0.25 0.39 — — 0.3 245.5 219.2 89.3 32.8 94.1 37.7

[0140] Referring to Table 2, it was confirmed that Examples 1 to 8 to which the positive electrode active material including the metal oxide in the form of a single particle on which a Co coating layer was formed had better electrochemical performance such as increased discharge capacity and improved life performance than Comparative Examples 1 and 2 having no Co coating layer formed thereon. In particular, it was found that the positive electrode active material according to Example 4 manufactured under the conditions of a heat treatment temperature of a coating layer formation process of 660° C. had the best characteristics of a discharge capacity at 0.2 C, lifespan, and a high-temperature resistance increase rate.

Experimental Example 4: Positive Electrode Active Material Structure Analysis

[0141] FIG. 1 is an SEM image measured at 20,000× magnification for a positive electrode active material prepared according to Example 4, and FIG. 2 is an SEM image measured at 20,000× magnification for a positive electrode active material prepared according to Comparative Example 3.

[0142] It was found that the positive electrode active material prepared according to the examples had a wave-patterned shape on the surface of single particles, as confirmed in FIG. 1. In this case, electrochemical performance was the best.

[0143] In addition, referring to FIG. 2, when the Co coating layer was formed by a wet method, it was confirmed that a stripe-shaped pattern was not included on the surface.

[0144] Next, FIG. 3 is an SEM image measured at 120,000× magnification for a positive electrode active material prepared according to Example 4.

[0145] In addition, a distance between protruding portions and a height of nine of the stripe shapes formed in the coating layer of the positive electrode active material formed according to Example 4 were measured and are shown in the following Table 3.

TABLE-US-00003 TABLE 3 Length in short Length in long side direction side direction of protruding of protruding portion (A)) portion (B) [nm] [nm] B/A ratio No 1 212.78 1127.65 5.3 No 2 174.72 1302.43 7.5 No 3 147.23 1385.62 9.4 No 4 169.48 912.78 5.4 No 5 168.33 966.86 5.7 No 6 146.44 886.3 6.1 No 7 126.15 705.34 5.6 No 8 157.24 1111.7 7.1 No 9 149.65 890.04 5.9

[0146] Referring to FIG. 3, it was found that the coating layer positioned on the surface of the positive electrode active material prepared according to Example 4 included a plurality of stripe shapes. In addition, referring to Table 4, it was confirmed that the length (B) in a length direction of the protruding portions forming the stripe shape and the length (A) in a width direction forming the stripe shape satisfied the range of Equation 1.

[0147] FIG. 4 is an SEM image measured at 60,000× magnification for a vertical cross section after processing a cross section in a length direction (direction B in FIG. 3) in the positive electrode active material prepared according to Example 4 using a focused ion beam (FIB), and FIG. 5 is schematically shown in order to describe a distance and a height between protruding portions forming a stripe shape of area A and area B in FIG. 4.

[0148] In addition, the images of FIG. 4 for the sample shown in Table 4 were taken, and then a distance (C) between the protruding portions of the areas corresponding to area A and area B and a height (D) of the protruding portions were measured and are shown in the following Table 4.

TABLE-US-00004 TABLE 4 Area A Area B Distance (C) Height (D) Distance (C) Height (D) between of between of protruding protruding protruding protruding portions portion D/C portions (C) portion D/C [nm] [nm] ratio [nm] [nm] ratio No 1 27.7 45.56 3.1 32.53 32.89 2.6 No 2 22.56 47.66 2.8 32.12 32.22 2.3 No 3 25.8 47.9 2.9 24.68 24.2 3.2 No 4 27.7 46.31 2.6 42.12 29.15 2.4 No 5 62.94 54.64 0.9 2.6 No 6 27.7 63.45 3.7 No 7 36.45 51.6 1.6 No 8 38.86 58.18 2.0 No 9 20.98 50.68 2.8

[0149] Referring to Table 4, in area A, the distance (C) between the protruding portions was 62.94 nm at 20.98 nm and the height (D) of the protruding portion was 63.45 nm at 45.56 nm. In area B, the distance (C) between the protruding portions was 42.12 nm at 24.68 nm and the height (D) of the protruding portion was 32.89 at 24.20. That is, it was confirmed that an average distance of the protruding portions forming a stripe shape formed on the coating layer in the positive electrode active material according to an exemplary embodiment was 20 nm or more, and an average height was 22 nm or more.

[0150] In addition, it was confirmed that the distance (C) between the protruding portions forming the stripe shape and the height (D) of the protruding portion forming the stripe shape satisfied the range of Equation 2.

[0151] Next, FIG. 6 is an SEM image measured at 5,000× magnification for a positive electrode active material prepared according to Example 8. That is, it relates to a positive electrode active material prepared in a bimodal form.

[0152] Referring to FIG. 6, it was confirmed that polycrystal having a large particle diameter and single particles having a small particle diameter were uniformly mixed.

Experimental Example 5: HPPC Evaluation

[0153] HPPC resistance was measured by the following method, using the positive electrode active materials prepared according to Example 4 and Comparative Example 3, and is shown in FIG. 7.

<Manufacture of Cell>

[0154] 1. Manufacture of 2032 coin cell (electrode density: 3.5 g/cc) [0155] 2. Resting the coin cell in a chamber at 25° C. for 10 hours (purpose: electrolyte solution impregnation and cell temperature synchronized with the temperature in the chamber)

<Cell HPPC Pattern>

[0156] 1. Confirmation of cell activation and capacity [0157] Charge: CC/CV {CC: 0.2C} {CV: 0.005C} [0158] Discharge: CC {CC: 0.2C} (reference [0159] capacity: 200 mAh/g) (cut off voltage: 2.5V-4.25V) [0160] 2. C-rate calculation based on the obtained capacity (capacity: about 4.5 mAh) (0.2C current: about 0.9 mA) [0161] 3. 0.2 C, 4.25 V full charge at the calculated C-rate (SOC 100% state) [0162] 4. {circle around (1)} 0.20, 30 min discharge (purpose: SOC-10%) [0163] {circle around (2)} 1C, 10 s discharge (purpose: measurement of resistance at target SOC) [0164] {circle around (3)} 0.2C, 50 s charge (purpose: compensation of lost SOC (capacity) in discharge at 1 C, 10 s) [0165] 5. Repeating the process of 4 a total of 9 times (SOC-10%)

<Calculation of HPPC Value>

[0166] Calculation of (90-10%) HPPC value for each SOC

(Initial voltage of {circle around (2)}-terminal voltage of {circle around (2)} in 4.)/(1C current of {circle around (2)})=(10 s Ω for each SOC)

[0167] Referring to FIG. 7, it was confirmed that the HPPC resistance value of the positive electrode active material prepared according to Example 4 on which the coating layer was formed by dry coating of Co was lower than the HPPC resistance value of the positive electrode active material prepared according to Comparative Example 3 on which Co was coated by a wet method.

[0168] HPPC resistance refers to a resistance value measured for each SOC of a battery, and the lower HPPC resistance, the less energy loss, and thus, the output of a battery may be improved. Since in the positive electrode active material of the present exemplary embodiment, the above HPPC resistance value was shown to be low, as confirmed in FIG. 7, it was found that it had excellent charging speed and output, and thus, was confirmed to be appropriate for use as a battery for an electric vehicle.

[0169] The positive electrode active material according to an exemplary embodiment has a single particle form, and has a modified surface structure so that the coating layer including a stripe shape is positioned on the surface, thereby implementing a positive electrode active material having a discharge capacity, room temperature resistance, a high-temperature lifespan, and high-temperature

resistance characteristics which are all improved, and thus, has a very advantageous effect.

[0170] The present disclosure is not limited by the above exemplary embodiments and may be manufactured in various forms different from each other, and it may be understood that a person with ordinary skill in the art to which the present disclosure pertains may carry out the present disclosure in another specific form without modifying the technical idea or essential feature of the present disclosure. Therefore, it should be understood that the exemplary embodiments described above are illustrative and are not restrictive in all aspects.

Claims

1. A positive electrode active material for a lithium secondary battery comprising: a metal oxide in the form of a single particle; and a coating layer positioned on a surface of the metal oxide, wherein the coating layer includes a plurality of stripe shapes including a protruding portion based on a cross section in a width direction.
2. The positive electrode active material for a lithium secondary battery of claim 1, wherein: the protruding portion has an average length in a short side direction of 110 nm or more.
3. The positive electrode active material for a lithium secondary battery of claim 1, wherein: the protruding portion has an average length in a long side direction of 650 nm or more.
4. The positive electrode active material for a lithium secondary battery of claim 1, wherein: an average distance between the protruding portions is in a range of 10 nm to 60 nm.
5. The positive electrode active material for a lithium secondary battery of claim 1, wherein: an average height of the protruding portion is in a range of 18 nm to 75 nm.
6. The positive electrode active material for a lithium secondary battery of claim 1, wherein: the stripe shape satisfies the following Equation 1: $4 < B / A < 11$ [Equation1] wherein A is a length in a short side direction of the protruding portion forming the stripe shape, and B is a length in a long side direction of the protruding portion forming the stripe shape.
7. The positive electrode active material for a lithium secondary battery of claim 1, wherein: the stripe shape satisfies the following Equation 2: $0.1 < D / C < 8$ [Equation2] wherein C is a distance between the protruding portions forming the stripe shape, and D is a height of the protruding portion forming the stripe shape.
8. The positive electrode active material for a lithium secondary battery of claim 1, wherein: the positive electrode active material has a Li/Ni cation mixing ratio of 1.5% or less.
9. The positive electrode active material for a lithium secondary battery of claim 1, wherein: the coating layer includes at least one of Co, Al, W, V, Ti, Nb, Ce, B, and P.
10. The positive electrode active material for a lithium secondary battery of claim 9, wherein: a content of the element included in the coating layer is in a range of 0.5 mol % to 3.5 mol % based on the entire coating layer.
11. The positive electrode active material for a lithium secondary battery of claim 1, wherein: the metal oxide includes nickel, cobalt, and manganese, and a content of the nickel is higher than the sum of contents of the cobalt and the manganese in the entire metal oxide.
12. The positive electrode active material for a lithium secondary battery of claim 11, wherein: the metal oxide further includes a doping element, and the doping element includes at least one of Al, Zr, Nb, Mo, W, Ti, Ce, Mg, B, P, V, Sr, and B.
13. The positive electrode active material for a lithium secondary battery of claim 1, wherein: a crystalline size of the metal oxide is 200 nm or more.
14. The positive electrode active material for a lithium secondary battery of claim 1, wherein: the positive electrode active material further includes a positive electrode active material including a metal oxide in the form of a secondary particle formed by agglomeration of primary particles.
15. The positive electrode active material for a lithium secondary battery of claim 14, wherein: an

average particle diameter (D50) of the positive electrode active material including the metal oxide in the form of a secondary particle is larger than an average particle diameter (D50) of the positive electrode active material including the metal oxide in the form of a single particle.

16. The positive electrode active material for a lithium secondary battery of claim 14, wherein: the metal oxide in the form of a secondary particle includes nickel, cobalt, and manganese, a content of the nickel is higher than the sum of contents of the cobalt and the manganese in the entire metal oxide in the form of a secondary particle.

17. The positive electrode active material for a lithium secondary battery of claim 16, wherein: the metal oxide in the form of a secondary particle further includes a doping element, and the doping element includes at least one of Al, Zr, Nb, Mo, W, Ti, Ce, Mg, B, P, V, Sr, and B.

18. The positive electrode active material for a lithium secondary battery of claim 14, wherein: compositions of the positive electrode active material including the metal oxides in the form of a single particle and in the form of a secondary particle are identical to or different from each other.

19. A lithium secondary battery comprising: a positive electrode including the positive electrode active material of claim 1; a negative electrode; and an electrolyte.
